

EXAMINATION 3 | CHEM 2313 | FALL 2019

Write your final answers in the boxes provided below. Additional paper has been provided at the end of the exam for scratch work along with supplemental tables. Only pencil, paper, a model kit, and one 8.5" × 11" sheet are permitted. Write all answers legibly and draw all structures clearly with sensible bond angles. Submit your crib sheet with the exam.

Name

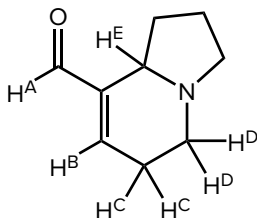
GT ID Number

Section	Total Points
I. Multiple-choice Questions	30
II. Predict the Products	20
III. Short-answer Questions	20
IV. Synthetic Planning	10
V. Reaction Mechanism	20

I. Multiple-Choice Questions

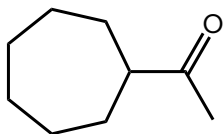
Choose the best answer among the choices provided. Circle your choice *and* write its letter in the box next to each question.

1. Of the labeled sets of hydrogens in the compound below, which set is the *most* acidic?



- A. H^A
- B. H^B
- C. H^C
- D. H^D
- E. H^E

2. Which of the following bases would result in formation and selective reaction (with an electrophile) of the *more substituted* enolate of the ketone below?

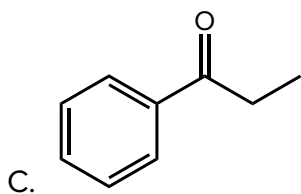
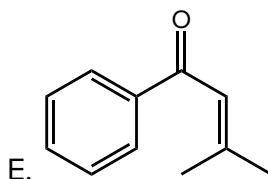
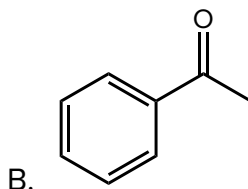
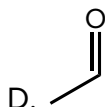
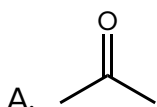
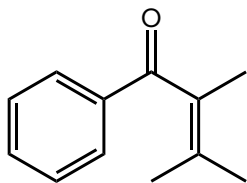


- A. Lithium diisopropyl amide (LDA, LiN(*i*-Pr)₂)
- B. Sodium acetate (NaOAc)
- C. Sodium hydroxide (NaOH)
- D. Sodium hydride (NaH)
- E. Lithium hexamethyldisilazide (LiN(SiMe₃)₂)

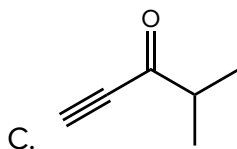
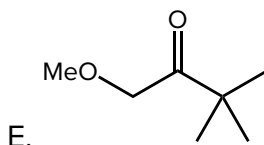
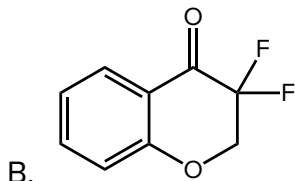
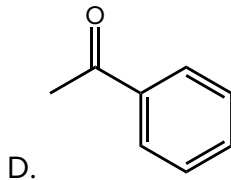
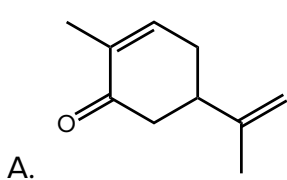
3. Which of the following statements concerning the malonic ester synthesis is *true*?

- A. A base at least as strong as lithium diisopropylamide (LDA) is required to deprotonate a malonic ester.
- B. The malonic ester synthesis is an alternative to the direct alkylation of acetone (i.e., it can lead to the same products).
- C. The decarboxylation step produces an alkane.
- D. Two alkylation steps can be carried out to produce a tertiary α -carbon.
- E. The alkoxy groups of the malonic ester must not match the alkoxide used.

4. Which of the following compounds is the *nucleophile* in the aldol condensation that produces the unsaturated ketone below?



5. Which of the following ketones *cannot* form an enol?

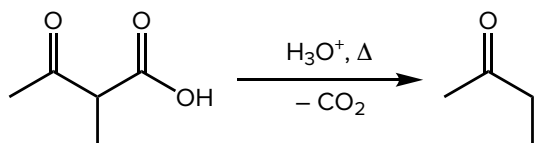


6. Which of the following elementary steps is the *first* irreversible step in the mechanism of the Claisen condensation?

☐

- A. Proton transfer from the ester to the alkoxide
- B. Nucleophilic addition of the ester enolate to another molecule of ester
- C. Beta-elimination of alkoxide
- D. Proton transfer from the β -ketoester to the alkoxide
- E. Protonation of the β -ketoester enolate by added acid

7. Which of the following is an intermediate in the thermal decarboxylation of a β -ketoacid under acidic conditions? An example is shown below.


☐

- A. enolate
- B. enol
- C. carboxylate
- D. carbocation
- E. alkoxide

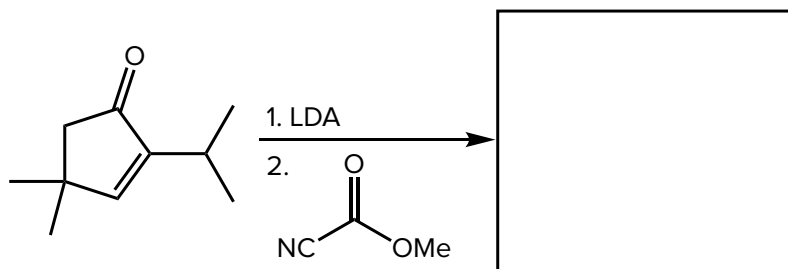
8. Which of the following groups is *NOT* electron withdrawing by resonance?

☐

- A. $-\text{CN}$
- B. $-\text{CO}_2\text{Me}$
- C. $-\text{SO}_3\text{H}$
- D. $-\text{NO}$
- E. $-\text{Cl}$

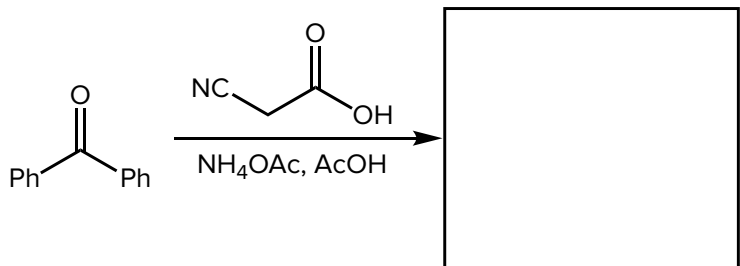
II. Predict the Products

Provide the major neutral organic product for each reaction or sequence of reactions below after appropriate workup. Clearly indicate the configurations of all stereocenters (except for those with random configurations; stereochemistry can be omitted at these).



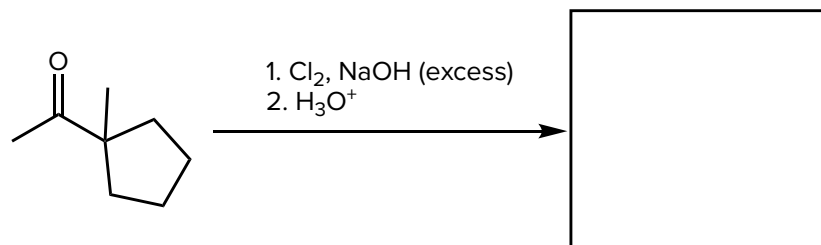
1.

LiCN is a byproduct.

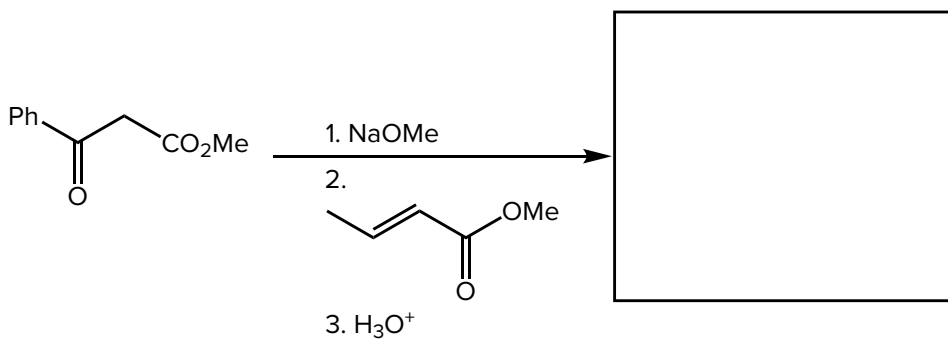


2.

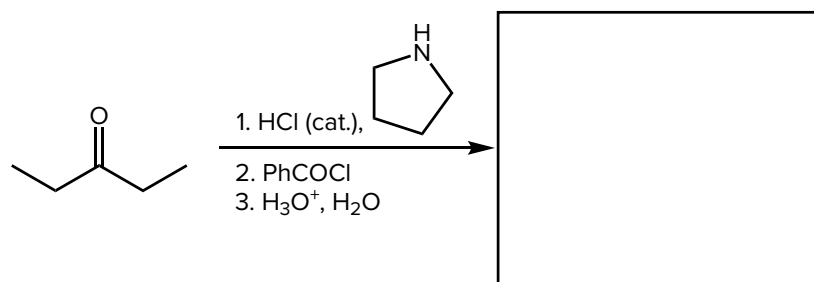
H₂O is a byproduct.



3.



4.

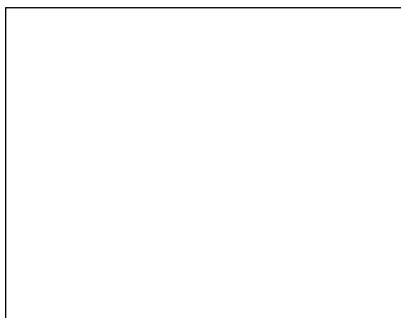


5.

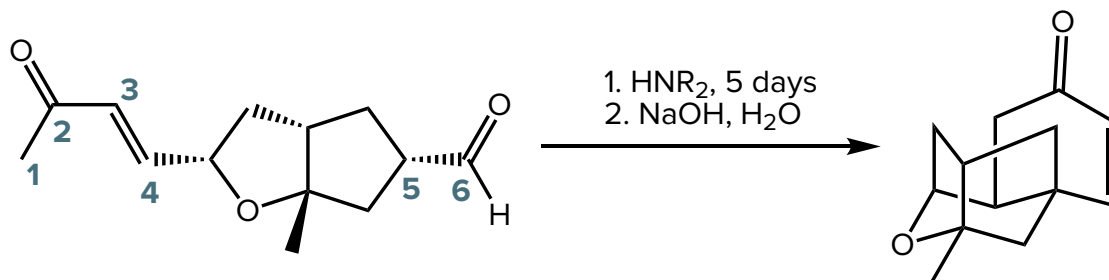
III. Short-answer Questions

For each question below, provide the requested structure, mechanism, or explanation. Ensure that your complete response fits within the box provided.

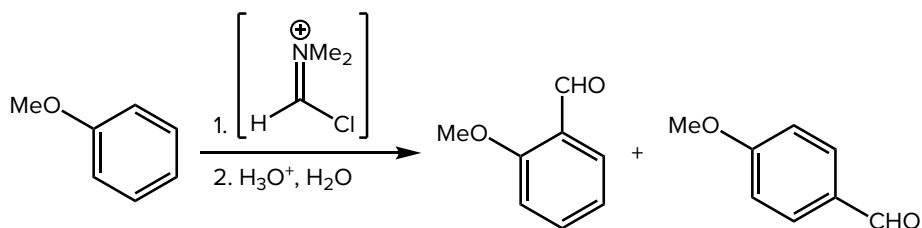
1. In the box at right, draw the structure of an ester that *cannot* react with itself to form a β -ketoester via Claisen condensation. Esters have the general formula R¹CO₂R².



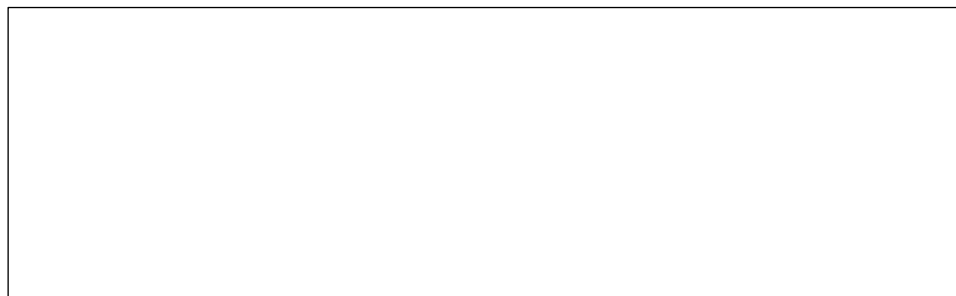
2. An example of the Robinson annulation is below. Examine the numbered carbons in the reactants and label the corresponding carbons in the product.



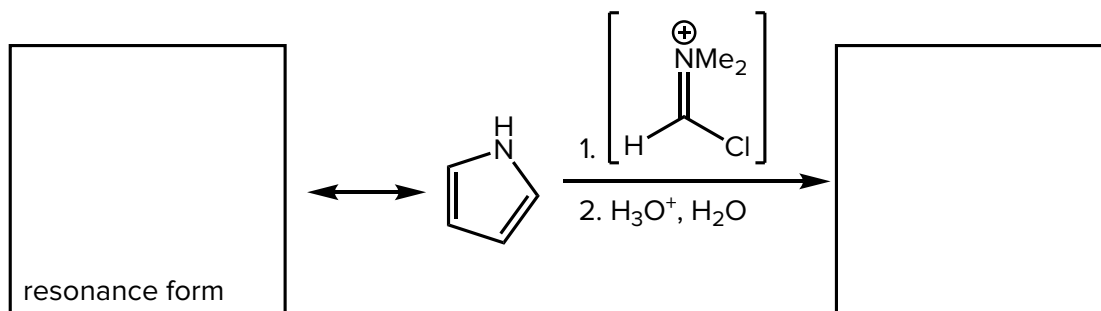
3. In the Vilsmeier reaction, a chloroiminium intermediate undergoes nucleophilic acyl substitution with an aromatic compound as nucleophile. An example is shown below. Dimethylammonium chloride ($\text{H}_2\text{NMe}_2^+ \text{Cl}^-$) is a byproduct of the reaction.



- a. In the box below, draw two resonance forms of anisole (PhOMe) that account for the two products observed. Note that anisole acts as a nucleophile in this reaction.

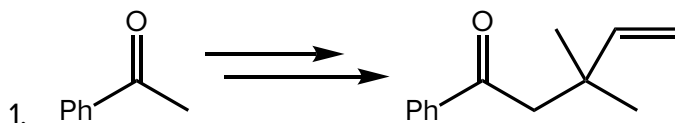


- b. Draw the major product when pyrrole is subjected to Vilsmeier conditions in the box below at right. In the box below at left, draw a resonance form of pyrrole that justifies formation of your product.

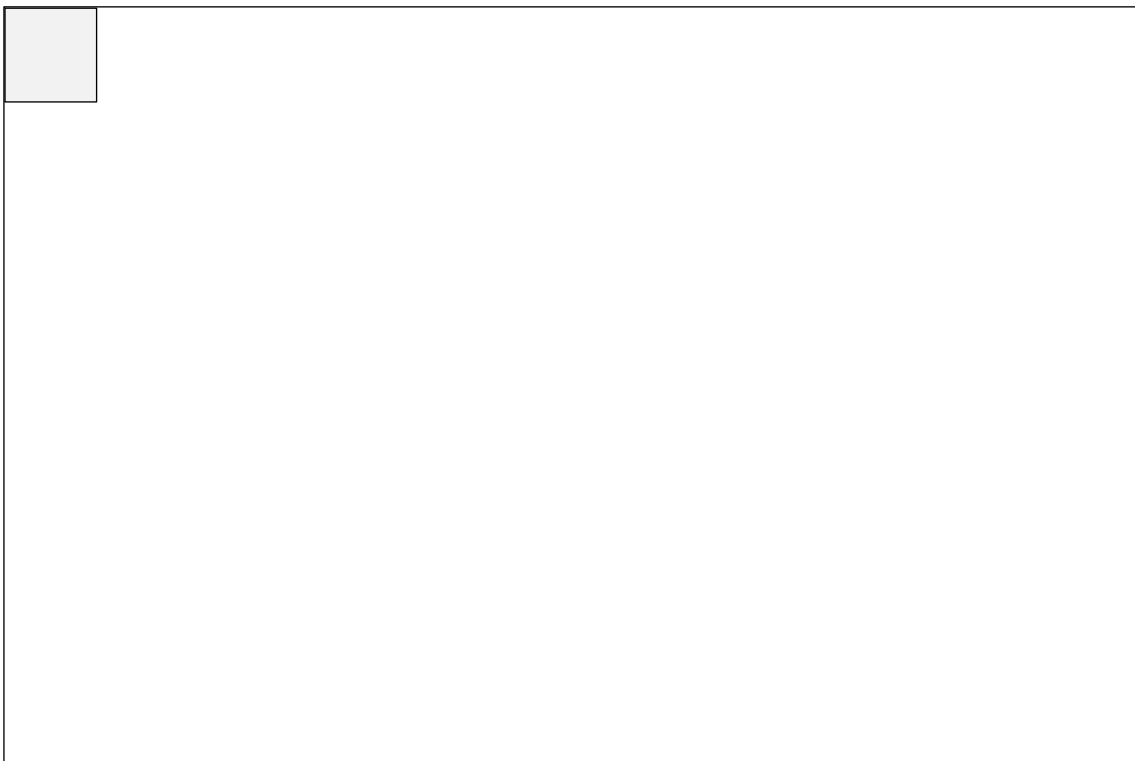
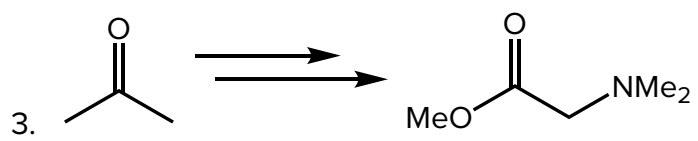
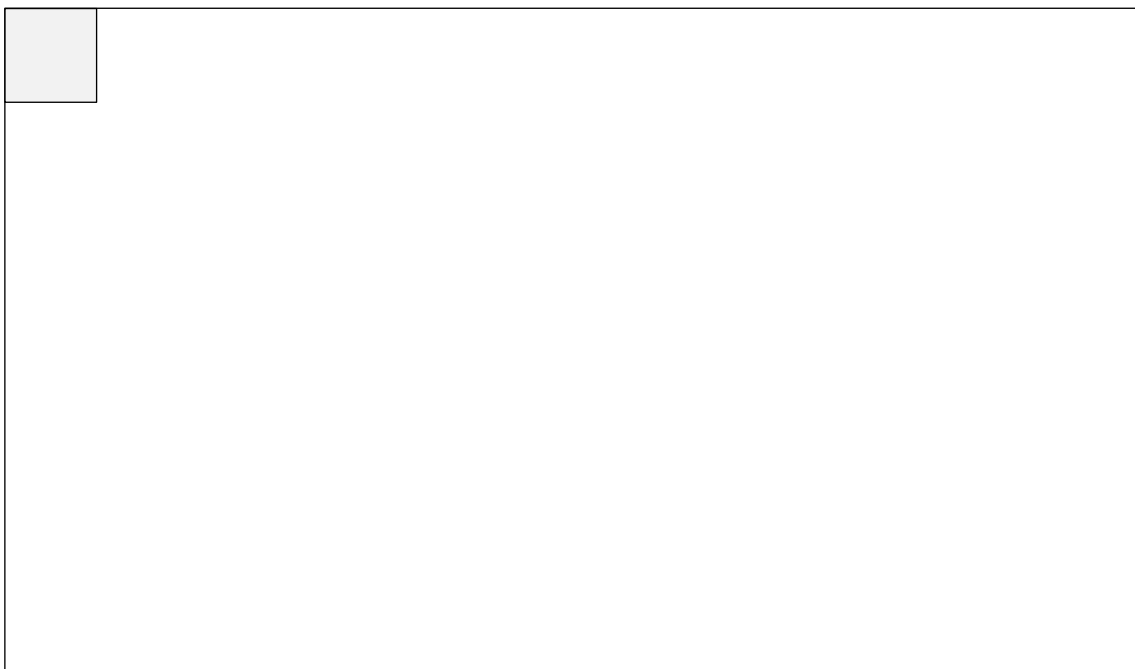
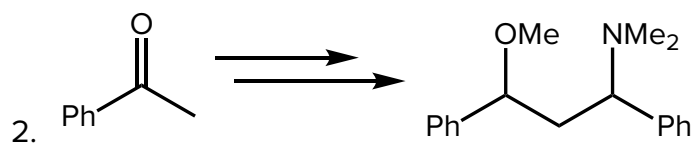


IV. Synthetic Planning

Of the three problems below, **choose two** and provide a sequence of reaction conditions that will accomplish the desired transformation. Include all isolated compounds, but do not draw curved arrows or any reactive intermediates. **Place an X in the shaded box next to the two problems you would like graded.**

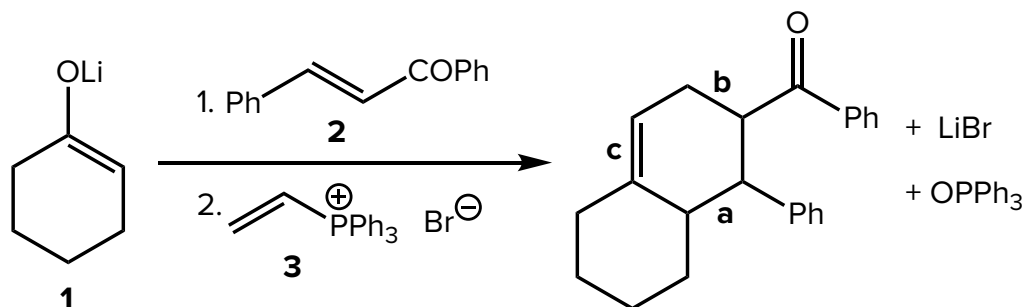


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V. Reaction Mechanism

Complete the problems below to describe the mechanism of the transformation shown.



- Bond **a** is the first carbon–carbon bond formed during the mechanism. Explain *in words* how bond **a** forms through the reaction of enolate **1** with α,β -unsaturated ketone **2**. In the box at right, draw the anionic intermediate **4** that follows this elementary step (omit Li⁺).



intermediate **4**

- Formation of bond **b** is next. Explain *in words* how bond **b** forms through the reaction of **4** with vinyl(triphenyl)phosphonium **3**. In the box at right, draw the ylide intermediate **5** that follows this elementary step.



intermediate **5**

- What type of reaction is involved in the conversion of **5** to the product through the formation of bond **c**? Either describe the mechanism in words or supply a named reaction; both will be awarded full credit.



pKa Chart

1

conjugate acid

conjugate base

sulfuric acid



-10

strongest acids

hydroiodic acid



-9

weakest bases

hydrobromic acid



-8

protonated ketone



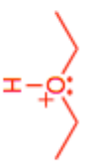
-7.3

hydrochloric acid



-7

protonated ether



-3.6

carbocations



-3

protonated alcohol



-2.4

hydronium ion



-1.7

2

conjugate acid

conjugate base

nitric acid



-1.3

tosic acid



-0.6

hydrofluoric acid



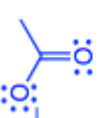
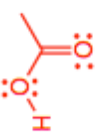
3.2

hydrogen nitride



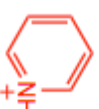
4.7

carboxylic acids



4.8

protonated pyridine



5.2

carbonic acid



(bicarbonate)

6.37

hydrogen sulfide

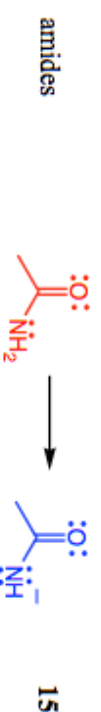
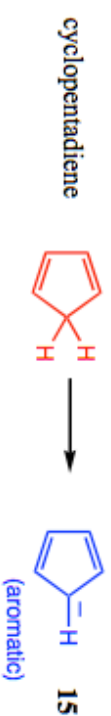
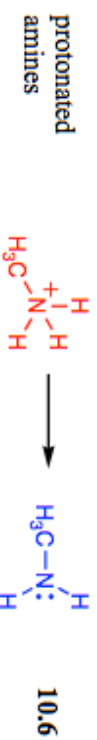
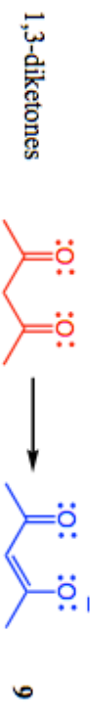


7

3

conjugate acid

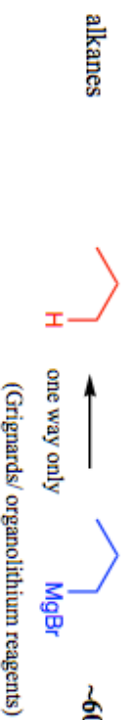
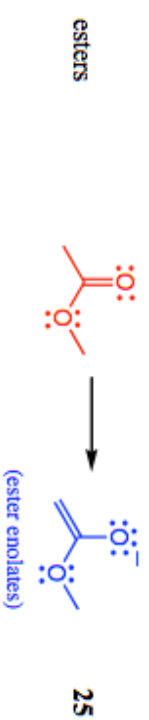
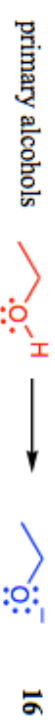
conjugate base



4

conjugate acid

conjugate base



weakest acids

strongest bases

(Grignards/ organolithium reagents)

Abbreviations in Organic Chemical Structures

Alkyl Groups

Abbreviation	Structure
Bu	
Cy	
Et	
i-Bu	
i-Pr	
Me	
t-Bu	

Ns	
Tf	
Ts	

Arenes

Abbreviation	Structure
Bn	
Ph	

Heteroatom-containing Groups

Abbreviation	Structure
Ac	
Bz	
Ms	
Nf	

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